

2(a)	It is a region of space where a body experiences a gravitational force by virtue of its mass.
2(b)(i)	As the diameters of the stars is very much smaller as compared to that of the distance between them, the stars can be considered as a point mass and thus Newton's law of gravitation may be used.
2(b)(ii)	$F = \frac{GMm}{r^2}$ $= \frac{(6.67 \times 10^{-11})(2.0 \times 10^{30})(2.4 \times 10^{29})}{(4.0 \times 10^{16})^2}$ $= 2.0 \times 10^{16} \text{ N}$ <u>Note:</u> - The distance $4.0 \times 10^{13} \text{ km}$ is between the 2 stars are measured from their centres. Hence there is no need to use the diameters. - This distance must be converted into metres.
2(c)	$a = \frac{2.0 \times 10^{16}}{2.0 \times 10^{30}}$ $= 1.0 \times 10^{-14} \text{ m s}^{-2}$ Although the force is large, it is insignificant as compared to the mass of the Sun. As such, the acceleration is insignificant and thus the Sun's motion is not affected.
3(a)(i)	aV